8.6 Miniature Circuit Breaker

MCB is an electro mechanical device which guards an electrical circuit from an over current that may effect from short circuit, overload or imperfect design. The structure of miniature circuit breaker is shown in figure 8.8.

- Normally work with currents below 100 amperes.
- They are the type of low voltage circuit breaker that is typically found in the fuse box of a residential or commercial building.



Fig. 8.8 **Circuit breaker**

8.6.1 Types of miniature circuit breaker

1. Single pole miniature circuit breaker

i. Single–pole circuit breakers are typically used for electric circuits with low power requirements that only use one live conductor, such as those providing power for lamps or general purpose electric outlets.

ii. A single–pole low voltage circuit breaker occupies only one space in a fuse box.

2. Double–pole miniature circuit breaker

i. It is used for devices that operate with two live conductors, such as domestic air conditioner models.

ii. When a double–pole circuit breaker is tripped, both lines are disconnected even if only one of them was involved in the electrical fault.

iii. A double–pole low voltage circuit breaker occupies two spaces in a fuse box.

3. Triple–pole miniature circuit breaker

i. It is used to protect electric equipment that works with three-phase power.

ii. The electric motors of elevators and large air conditioning units typically work with three-phase power and are protected with triple pole low voltage circuit breakers.

iii. A triple-pole MCB low voltage circuit breaker occupies three spaces in a fuse box.

8.7 Moulded Case Circuit Breaker (MCCB)

This circuit breaker is an electro mechanical device which guards a circuit from short circuit and over current. The short circuit and over current protection for circuits ranging from 63 Amps-3000 Amps. The primary functions of MCCB

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